Tiago Santos

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Education

Instituto Superior Técnico — Lisbon, Portugal
B.Sc. in Computer Science and Engineering — *Sept. 2025 – Expected June 2028*

- **Relevant coursework:** Object-Oriented Programming, Computer Architecture, Operating Systems, Algorithms and Data Structures

Technical Skills

Languages: C, Java, Lua, TypeScript, JavaScript, Python, RISC-V Assembly
Frameworks and Databases: React, MySQL
Tools and Systems: Git, Linux, Make, Valgrind, Logisim, POSIX, pthreads

Experience

Yoda Things — Lisbon, Portugal
Co-Founder and Software Developer — *June 2024 – Present*

- Co-founded and commercialized Lua scripting solutions for multiplayer game servers, serving 30+ paying clients.
- Designed and maintained 5+ event-driven Lua subsystems for gameplay state, player inventories, and in-game economy persistence backed by MySQL.
- Developed and maintained server-side scripts for live multiplayer environments, including the Saga server, which supported up to 100 concurrent players.
- Profiled scripts, optimized execution paths, and added defensive error handling around server workloads.
- Diagnosed and fixed issues across server logic, persistence, and client-facing behavior.

Selected Projects

CloudIST – POSIX Data Center Simulator | *Academic team project (2 students), C, POSIX* — In progress

- Building a data-center simulator for server provisioning, virtual-machine reservations, resource validation, and concurrent workload execution.
- Implemented process execution, operating-system resource controls, and synchronization mechanisms for safe concurrent workloads.
- Wrote integration tests covering process execution, concurrency behavior, resource consistency, and runtime cleanup.

MiniTransformerGPT – Transformer LM and Custom Processor | *RISC-V Assembly, Logisim* — [Code]

- Implemented a bare-metal Transformer inference engine from scratch in RISC-V Assembly, including self-attention, matrix/vector operations, and token selection.
- Designed a custom 16-bit processor in Logisim with dedicated vector hardware for matrix operations and wrote regression tests for core computations.

In-Memory Logistics and Billing Engine | *C, Make, Valgrind* — [Code]

- Built a command-line invoicing and inventory engine supporting product records, transaction tracking, and EAN validation.
- Implemented in-memory data structures for 10,000+ product records and used Valgrind and stress testing to eliminate leaks and invalid accesses.

ERPlatform – Retail Platform Domain Model | *Academic team project (2 students), Java, UML* — In progress

- Designing an object-oriented retail platform in Java (companies, products, carts, orders, notifications), with UML models for domain relationships and lifecycle transitions.

Additional Information

Other projects: Portfolio website (React, TypeScript, Vite) ScrabblePY (Python)
Languages: Portuguese (Native), English (Fluent)